

FagusED · A-Level 9702  
1 Physical quantities and units

What is the central idea in Physical quantities and units?

A physical quantity combines a number and a unit. Base SI quantities include mass, length, time and current; derived units are products of base units.

9702-01-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
1 Physical quantities and units

A resistance is calculated from  $6.0 \pm 0.2$  V and  $2.0 \pm 0.1$  A. Determine R and its approximate absolute uncertainty.

$R = V/I = 3.0 \Omega$ . Percentage uncertainty =  $0.2/6.0 + 0.1/2.0 = 0.0833$ . Absolute uncertainty  $\approx 0.25 \Omega$ .

9702-01-P

QUESTION + ANSWER

FagusED · A-Level 9702  
2 Kinematics

A trolley accelerates uniformly from rest to  $9 \text{ m s}^{-1}$  over 18 m. Find acceleration and travel time.

$v^2 = u^2 + 2as$ , giving  $a = 81/36 = 2.25 \text{ m s}^{-2}$ ;  $t = (v - u)/a = 4.0 \text{ s}$ .

9702-02-P

QUESTION + ANSWER

FagusED · A-Level 9702  
3 Dynamics

Which mechanism, condition or distinction matters in Dynamics?

Momentum is a vector and is conserved in an isolated interaction. During a collision, increasing contact time reduces average force for a fixed change in momentum.

9702-03-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
4 Forces, density and pressure

What is the central idea in Forces, density and pressure?

Force diagrams must show forces on one chosen body. Resolve vectors before finding the resultant or equilibrium.

9702-04-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
1 Physical quantities and units

Which mechanism, condition or distinction matters in Physical quantities and units?

Dimensional homogeneity checks equations, while percentage uncertainties add for products or quotients. Precision describes repeatability; accuracy concerns closeness to the true value.

9702-01-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
2 Kinematics

Which mechanism, condition or distinction matters in Kinematics?

The constant-acceleration equations apply only where acceleration remains uniform.

9702-02-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
2 Kinematics

What is the gradient of a velocity-time graph?

A Displacement  
B Acceleration  
C Distance  
D Power

B. Acceleration  
Why: Displacement is a vector change in position; velocity is its rate of change.

9702-02-M

QUESTION + ANSWER

FagusED · A-Level 9702  
3 Dynamics

A 0.6 kg cart at  $4 \text{ m s}^{-1}$  collides and sticks to a stationary 0.4 kg cart. Find their common velocity and state the condition for conservation.

Initial momentum  $0.6 \times 4 = 2.4 \text{ kg m s}^{-1}$ ; common velocity  $2.4/(0.6+0.4) = 2.4 \text{ m s}^{-1}$ ; negligible external resultant impulse.

9702-03-P

QUESTION + ANSWER

FagusED · A-Level 9702  
1 Physical quantities and units

Which quantity has SI base unit kg?

A Mass  
B Force  
C Pressure  
D Energy

A. Mass  
Why: A physical quantity combines a number and a unit.

9702-01-M

QUESTION + ANSWER

FagusED · A-Level 9702  
2 Kinematics

What is the central idea in Kinematics?

Displacement is a vector change in position; velocity is its rate of change. A displacement-time gradient is velocity and velocity-time gradient is acceleration; signed area under velocity-time is displacement.

9702-02-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
3 Dynamics

What is the central idea in Dynamics?

Newton's second law relates resultant force to the rate of change of momentum:  $F = \Delta p/\Delta t$ , reducing to  $ma$  for constant mass.

9702-03-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
3 Dynamics

For an isolated collision, which total is conserved?

A Kinetic energy always  
B Temperature  
C Momentum  
D Speed of each body

C. Momentum  
Why: Newton's second law relates resultant force to the rate of change of momentum:  $F = \Delta p/\Delta t$ , reducing to  $ma$  for constant mass.

9702-03-M

QUESTION + ANSWER

FagusED · A-Level 9702  
4 Forces, density and pressure

Which mechanism, condition or distinction matters in Forces, density and pressure?

The moment about a pivot is force times perpendicular lever arm. Density is mass/volume and hydrostatic pressure change  $\Delta p = \rho g \Delta h$ ; pressure acts normally to a surface.

9702-04-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
4 Forces, density and pressure

Pressure increase at depth  $h$  in a liquid is proportional to what?

- A  $1/h$
- B  $h^2$  only
- C Surface area
- D  $\rho gh$

D.  $\rho gh$

Why: Force diagrams must show forces on one chosen body.

9702-04-M

QUESTION + ANSWER

FagusED · A-Level 9702  
4 Forces, density and pressure

A water tank is filled to depth 1.8 m. Find the pressure increase from surface to bottom, using  $\rho = 1000 \text{ kg m}^{-3}$  and  $g = 9.81 \text{ m s}^{-2}$ .

$$\Delta p = \rho gh = 1000 \times 9.81 \times 1.8 = 1.77 \times 10^4 \text{ Pa (3 s.f.)}$$

9702-04-P

QUESTION + ANSWER

FagusED · A-Level 9702  
5 Work, energy and power

Which mechanism, condition or distinction matters in Work, energy and power?

Power measures energy transferred per unit time. For a real process, conservation applies to all stores even when useful efficiency is below 100%.

9702-05-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
5 Work, energy and power

Which expression gives kinetic energy?

- A  $\frac{1}{2}mv^2$
- B  $mv$
- C  $mgh$
- D  $F/t$

A.  $\frac{1}{2}mv^2$

Why: Work is energy transfer by a force through displacement in its direction:  $W = Fs \cos \theta$ .

9702-05-M

QUESTION + ANSWER

FagusED · A-Level 9702  
5 Work, energy and power

What is the central idea in Work, energy and power?

Work is energy transfer by a force through displacement in its direction:  $W = Fs \cos \theta$ . Kinetic energy is  $\frac{1}{2}mv^2$ ; gravitational energy change near Earth is  $mg\Delta h$ .

9702-05-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
6 Deformation of solids

What is the central idea in Deformation of solids?

Stress = force/cross-sectional area; strain = extension/original length. Young modulus  $E$  = stress/strain in a linear elastic region.

9702-06-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
6 Deformation of solids

Which mechanism, condition or distinction matters in Deformation of solids?

The gradient of a force-extension graph is spring constant; its area gives work done. Elastic deformation is reversible on unloading, whereas plastic deformation leaves a permanent extension.

9702-06-C2

QUESTION + ANSWER

FagusED · A-Level 9702  
5 Work, energy and power

A motor lifts a 30 kg mass through 2.0 m in 5.0 s while drawing 150 W. Calculate efficiency.

$$\text{Useful power} = mgh/t = 30 \times 9.81 \times 2/5 = 117.7 \text{ W. Efficiency} = 117.7/150 = 0.785, \text{ or } 78.5\%.$$

9702-05-P

QUESTION + ANSWER

FagusED · A-Level 9702  
6 Deformation of solids

A spring has  $k = 80 \text{ N m}^{-1}$  and extends by 0.050 m. Find force and elastic energy stored.

$$F = kx = 4.0 \text{ N. Energy} = \frac{1}{2}kx^2 = 0.10 \text{ J.}$$

9702-06-P

QUESTION + ANSWER

FagusED · A-Level 9702  
7 Waves

What is the central idea in Waves?

A progressive wave transfers energy without a net transport of matter. Its speed is  $v = f\lambda$ . Transverse displacement is perpendicular to propagation; longitudinal displacement is parallel.

9702-07-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
6 Deformation of solids

Which ratio gives Young modulus?

- A Strain/stress
- B Stress/strain
- C Force/extension only
- D Extension/length only

B. Stress/strain

Why: Stress = force/cross-sectional area; strain = extension/original length.

9702-06-M

QUESTION + ANSWER

FagusED · A-Level 9702  
7 Waves

Which mechanism, condition or distinction matters in Waves?

Intensity scales as amplitude squared for otherwise comparable waves. Superposition adds simultaneous displacements algebraically.

9702-07-C2

QUESTION + ANSWER

How is wave speed related to frequency and wavelength?

A  $v=f/\lambda$

B  $v=\lambda/f$

C  $v=f\lambda$

D  $v=f+\lambda$

C.  $v=f\lambda$

Why: A progressive wave transfers energy without a net transport of matter.

9702-07-M

QUESTION + ANSWER

Two slits are 0.30 mm apart and 1.5 m from a screen. Adjacent bright fringes are 3.0 mm apart. Find wavelength.

Use  $x = \lambda D/a$ , so  $\lambda = xa/D = (3.0 \times 10^{-3})(3.0 \times 10^{-4})/1.5 = 6.0 \times 10^{-7}$  m.

9702-08-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Electricity?

Resistance  $R = V/I$ ; an ohmic conductor at constant temperature has a linear I-V graph through the origin. Electrical power  $P = VI = I^2R$ . Resistivity depends on material and temperature.

9702-09-C2

QUESTION + ANSWER

What is the central idea in D.C. circuits?

Kirchhoff's first law conserves charge at a junction; the second conserves energy around a loop.

9702-10-C1

QUESTION + ANSWER

A cell has  $\mathcal{E} = 9.0$  V and internal resistance  $0.50 \Omega$ ; external resistance is  $4.0 \Omega$ . Find current and terminal p.d.

$I = \mathcal{E}/(R+r) = 9/4.5 = 2.0$  A;  $V = IR = 8.0$  V.

9702-10-P

QUESTION + ANSWER

A wave travels at  $340 \text{ m s}^{-1}$  and has frequency 680 Hz. Find wavelength and period.

$\lambda = v/f = 0.50$  m;  $T = 1/f = 1.47 \times 10^{-3}$  s.

9702-07-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Superposition?

Diffraction spreads waves after an aperture, especially where width is comparable with wavelength. Stationary waves form nodes and antinodes from counter-propagating waves.

9702-08-C2

QUESTION + ANSWER

What is the central idea in Electricity?

Electric current is charge flow rate  $I = \Delta Q/\Delta t$ . Potential difference is energy transferred per unit charge,  $V = W/Q$ .

9702-09-C1

QUESTION + ANSWER

A wire of length 4.0 m and area  $2.0 \times 10^{-6} \text{ m}^2$  has resistance  $0.10 \Omega$ . Determine resistivity.

$\rho = RA/L = 0.10 \times 2.0 \times 10^{-6}/4.0 = 5.0 \times 10^{-8} \Omega \text{ m}$ .

9702-09-P

QUESTION + ANSWER

How is terminal p.d. of a discharging cell related to e.m.f.?

A  $V=\mathcal{E}+Ir$

B  $V=\mathcal{E}-Ir$

C  $V=Ir-\mathcal{E}$

D  $V=\mathcal{E}/I$

B.  $V=\mathcal{E}-Ir$

Why: Kirchhoff's first law conserves charge at a junction; the second conserves energy around a loop.

9702-10-M

QUESTION + ANSWER

What is the central idea in Superposition?

Interference needs coherent sources with a stable phase difference. Constructive interference arises at path differences  $n\lambda$  and destructive at  $(n + \frac{1}{2})\lambda$  for in-phase sources.

9702-08-C1

QUESTION + ANSWER

For coherent in-phase waves, which path difference yields a bright fringe?

A  $\lambda/2$

B  $3\lambda/2$

C  $\lambda/4$

D  $\lambda$

D.  $\lambda$

Why: Interference needs coherent sources with a stable phase difference.

9702-08-M

QUESTION + ANSWER

For a resistor at fixed conditions, which expression gives power?

A  $I^2R$

B  $I/R$

C  $R/V$

D  $Q/t^2$

A.  $I^2R$

Why: Electric current is charge flow rate  $I = \Delta Q/\Delta t$ .

9702-09-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in D.C. circuits?

In series, current is shared and potential differences add; in parallel, potential difference is shared and currents add. A source with internal resistance  $r$  has terminal p.d.  $V = \mathcal{E} - Ir$  during discharge.

9702-10-C2

QUESTION + ANSWER

What is the central idea in Particle physics?

Hadrons are made of quarks: baryons contain three quarks and mesons a quark-antiquark pair. Leptons, including electrons and neutrinos, are fundamental.

9702-11-C1

QUESTION + ANSWER

In  $\beta^-$  decay, a neutron becomes a proton, electron and electron antineutrino. Verify charge and lepton-number conservation.

Charge:  $0 \rightarrow +1 -1 +0 = 0$ . Electron lepton number:  $0 \rightarrow +1 -1 = 0$ . Baryon number remains +1.

9702-11-P

QUESTION + ANSWER

An object travels a horizontal circle of radius 0.80 m in 2.0 s. Determine angular speed and centripetal acceleration.

$\omega = 2\pi/T = \pi \text{ rad s}^{-1}$ ;  $a = \omega^2 r = 7.90 \text{ m s}^{-2}$ .

9702-12-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Gravitational fields?

The negative sign records an attractive bound state. Circular satellites satisfy  $GMm/r^2 = mv^2/r$ ; a geostationary orbit has Earth's rotational period.

9702-13-C2

QUESTION + ANSWER

What is the central idea in Temperature?

Temperature measures thermal state and determines direction of net thermal energy transfer. At absolute zero kelvin is zero, though particles still have quantum behaviour.

9702-14-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Particle physics?

Particle reactions must conserve charge, baryon number and lepton number; mass-energy may be converted according to  $E = mc^2$ .

9702-11-C2

QUESTION + ANSWER

What is the central idea in Motion in a circle?

Circular motion changes velocity direction even at constant speed. The centripetal acceleration  $v^2/r = \omega^2 r$  points toward the centre, and its resultant force is  $mv^2/r$ .

9702-12-C1

QUESTION + ANSWER

What direction is centripetal acceleration?

- A Tangential to motion
- B Away from centre
- C Vertical always
- D Towards centre

D. Towards centre  
Why: Circular motion changes velocity direction even at constant speed.

9702-12-M

QUESTION + ANSWER

How does gravitational field strength vary outside a spherical mass?

- A As  $1/r^2$
- B As  $r^2$
- C As  $1/r$
- D Constant with  $r$

A. As  $1/r^2$ 

Why: Gravitational field strength is force per unit mass,  $g = GM/r^2$  outside a spherical mass.

9702-13-M

QUESTION + ANSWER

What is the quark content of a proton?

- A udd
- B uuu
- C uud
- D ddd

C. uud

Why: Hadrons are made of quarks: baryons contain three quarks and mesons a quark-antiquark pair.

9702-11-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Motion in a circle?

Angular speed  $\omega = 2\pi/T$ . "Centripetal force" describes a resultant supplied by real forces, not an extra interaction.

9702-12-C2

QUESTION + ANSWER

What is the central idea in Gravitational fields?

Gravitational field strength is force per unit mass,  $g = GM/r^2$  outside a spherical mass. Potential is work per unit mass relative to infinity,  $\phi = -GM/r$ .

9702-13-C1

QUESTION + ANSWER

At radius  $r$ , orbital speed is  $v$ . Derive the new speed for a circular orbit of radius  $4r$  about the same central mass.

Equate  $GMm/r^2 = mv^2/r$  to give  $v = \sqrt{(GM/r)}$ . At  $4r$ , speed is  $v/2$ .

9702-13-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Temperature?

A thermometric property must vary predictably with temperature; a calibrated scale turns property readings into temperatures.  $\Delta T$  in kelvin equals  $\Delta T$  in  $^{\circ}\text{C}$ .

9702-14-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Ideal gases?

Use kelvin, pascals and cubic metres; a real gas approaches ideal behaviour at low pressure and high temperature.

9702-15-C2

QUESTION + ANSWER

A sealed rigid vessel at 300 K has  $p = 100 \text{ kPa}$ . Find  $p$  at 450 K and state the assumption.

At fixed volume and amount,  $p/T$  is constant.  $p_2 = 100 \times 450/300 = 150 \text{ kPa}$ . Assume ideal gas behaviour.

9702-15-P

QUESTION + ANSWER

A gas receives heat  $Q$  and does work  $W_{\text{out}}$ . What is  $\Delta U$ ?

- A  $Q + W_{\text{out}}$   
 B  $W_{\text{out}} - Q$   
 C  $-Q - W_{\text{out}}$   
 D  $Q - W_{\text{out}}$

D.  $Q - W_{\text{out}}$

Why: Internal energy is the total random microscopic kinetic and potential energy of particles.

9702-16-M

QUESTION + ANSWER

Which conversion to kelvin is closest for  $27^{\circ}\text{C}$ ?

- A 27 K  
 B 300 K  
 C 246 K  
 D 373 K

B. 300 K

Why: Temperature measures thermal state and determines direction of net thermal energy transfer.

9702-14-M

QUESTION + ANSWER

Which is the ideal-gas equation?

- A  $p/T = nRV$   
 B  $pV = nR/T$   
 C  $pV = nRT$   
 D  $p = nVRT$

C.  $pV = nRT$

Why: The ideal gas equation  $pV = nRT$  links macroscopic state variables.

9702-15-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Thermodynamics?

During melting at constant temperature, energy increases molecular potential energy. Specific heat capacity relates heating to  $mc\Delta T$ ; latent heat relates phase change to  $mL$ .

9702-16-C2

QUESTION + ANSWER

What is the central idea in Oscillations?

Simple harmonic motion has acceleration proportional to displacement and opposite in direction:  $a = -\omega^2 x$ . A mass-spring system has  $T = 2\pi\sqrt{m/k}$  when damping is negligible.

9702-17-C1

QUESTION + ANSWER

A gas warms from  $20^{\circ}\text{C}$  to  $80^{\circ}\text{C}$ . Give initial and final absolute temperatures and temperature rise in kelvin.

Initial = 293 K; final = 353 K; rise = 60 K.

9702-14-P

QUESTION + ANSWER

What is the central idea in Ideal gases?

The ideal gas equation  $pV = nRT$  links macroscopic state variables. In kinetic theory, pressure arises from collisions and mean translational kinetic energy per molecule is  $3kT/2$ .

9702-15-C1

QUESTION + ANSWER

What is the central idea in Thermodynamics?

Internal energy is the total random microscopic kinetic and potential energy of particles. First law:  $\Delta U = Q + W$  when  $W$  is work done on the system (check sign convention).

9702-16-C1

QUESTION + ANSWER

A gas receives 400 J of heating and does 150 J of work on its surroundings. Find  $\Delta U$  and explain the sign of the work term.

Work done on gas =  $-150 \text{ J}$ .  $\Delta U = Q + W = 400 - 150 = +250 \text{ J}$ ; expansion transfers energy out of the gas.

9702-16-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Oscillations?

At equilibrium speed is maximum; at extreme displacement speed is zero. Resonance occurs when driving frequency is close to natural frequency.

9702-17-C2

QUESTION + ANSWER

Which expression describes SHM acceleration?

A  $a = -\omega^2 x$

B  $a = +\omega^2 x$

C  $a = \omega x^2$

D  $a = 0$  throughout

A.  $a = -\omega^2 x$

Why: Simple harmonic motion has acceleration proportional to displacement and opposite in direction:  $a = -\omega^2 x$ .

9702-17-M

QUESTION + ANSWER

Parallel plates 5.0 mm apart have p.d. 300 V. Find field strength and work done moving an electron across the p.d.

$$E = V/d = 300/0.005 = 6.0 \times 10^4 \text{ V m}^{-1}; \text{ energy magnitude } eV = 1.60 \times 10^{-19} \times 300 = 4.8 \times 10^{-17} \text{ J.}$$

9702-18-P

QUESTION + ANSWER

For capacitance  $C$  at p.d.  $V$ , energy stored is what?

A  $CV^2$

B  $2CV^2$

C  $\frac{1}{2}CV^2$

D  $\frac{1}{2}C/V^2$

C.  $\frac{1}{2}CV^2$

Why: Capacitance  $C = Q/V$  measures charge stored per potential difference.

9702-19-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Magnetic fields?

Faraday's law relates induced e.m.f. to the rate of change of magnetic flux linkage, with Lenz's law setting the opposing direction.

9702-20-C2

QUESTION + ANSWER

A spring oscillator has mass 0.50 kg and  $k = 200 \text{ N m}^{-1}$ . Find period and describe the effect of quadrupling mass.

$$T = 2\pi\sqrt{(m/k)} = 2\pi\sqrt{(0.50/200)} \approx 0.314 \text{ s. Quadrupling } m \text{ doubles } T.$$

9702-17-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Electric fields?

Electric potential is energy per unit charge. A positive charge tends to move in the direction of decreasing potential when free.

9702-18-C2

QUESTION + ANSWER

What is the central idea in Capacitance?

Capacitance  $C = Q/V$  measures charge stored per potential difference. A capacitor stores energy  $E = \frac{1}{2}QV = \frac{1}{2}CV^2$ .

9702-19-C1

QUESTION + ANSWER

A  $220 \mu\text{F}$  capacitor discharges through  $10 \text{ k}\Omega$ . Find its time constant and fraction of voltage remaining after one time constant.

$$\tau = RC = 10\,000 \times 220 \times 10^{-6} = 2.2 \text{ s; fraction } e^{-1} \approx 0.368.$$

9702-19-P

QUESTION + ANSWER

Magnetic force on a moving charge at right angles to  $B$  is what?

A  $Bq/v$

B  $Bv/q$

C  $qv/B$

D  $Bqv$

D.  $Bqv$

Why: A current-carrying conductor in a magnetic field experiences  $F = BIL \sin \theta$ .

9702-20-M

QUESTION + ANSWER

What is the central idea in Electric fields?

Electric field strength  $E$  is force per unit positive charge. Between parallel plates, uniform  $E = V/d$ ; a point charge produces a radial field proportional to  $1/r^2$ .

9702-18-C1

QUESTION + ANSWER

Between uniform parallel plates,  $E$  equals what?

A  $Vd$

B  $V/d$

C  $d/V$

D  $V^2d$

B.  $V/d$

Why: Electric field strength  $E$  is force per unit positive charge.

9702-18-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Capacitance?

During resistor-capacitor discharge,  $V = V_0 e^{-(t/RC)}$ ; the time constant  $RC$  is the time for voltage to fall to  $1/e$  of its initial value.

9702-19-C2

QUESTION + ANSWER

What is the central idea in Magnetic fields?

A current-carrying conductor in a magnetic field experiences  $F = BIL \sin \theta$ . A moving charge experiences  $F = Bqv \sin \theta$ ; the force is perpendicular to velocity and does no work.

9702-20-C1

QUESTION + ANSWER

A 50-turn coil's flux per turn changes from 0.020 to 0.0050 Wb in 0.10 s. Find mean induced e.m.f. magnitude.

$$\Delta(N\Phi) = 50(0.020 - 0.0050) = 0.75 \text{ Wb-turn; } \text{abs}(\epsilon) = \Delta(N\Phi)/\Delta t = 7.5 \text{ V.}$$

9702-20-P

QUESTION + ANSWER

What is the central idea in Alternating currents?

An alternating voltage changes sign periodically. For a sinusoid  $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$ ; resistive power uses RMS current and voltage.

9702-21-C1

QUESTION + ANSWER

An ideal transformer has 1000 primary turns, 50 secondary turns and 240 V primary RMS. Determine secondary voltage and primary current if secondary current is 2.0 A.

$V_s = 240 \times 50/1000 = 12 \text{ V}$ ; equal powers:  $240 I_p = 12 \times 2.0$ ,  $I_p = 0.10 \text{ A}$ .

9702-21-P

QUESTION + ANSWER

What increases the maximum kinetic energy of photoelectrons above threshold?

- A Light intensity alone
- B Light frequency
- C Illuminated area alone
- D Exposure time alone

B. Light frequency  
Why: A photon has energy  $hf$  and momentum  $h/\lambda$ .

9702-22-M

QUESTION + ANSWER

A sample starts at 640 Bq and its half-life is 5.0 h. Find activity after 15 h, then state what one becquerel means.

15 h is three half-lives.  $A = 640/2^3 = 80 \text{ Bq}$ . One Bq is one nuclear decay per second.

9702-23-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Alternating currents?

An ideal transformer obeys  $V_s/V_p = N_s/N_p$  and conserves input/output power. Transmission at high voltage lowers current and  $I^2R$  losses for given power.

9702-21-C2

QUESTION + ANSWER

What is the central idea in Quantum physics?

A photon has energy  $hf$  and momentum  $h/\lambda$ . Photoelectric emission requires photon frequency above a material's threshold: maximum kinetic energy =  $hf - \text{work function}$ .

9702-22-C1

QUESTION + ANSWER

A metal's threshold frequency is  $5.0 \times 10^{14} \text{ Hz}$ . Light of  $8.0 \times 10^{14} \text{ Hz}$  falls on it. Find maximum kinetic energy,  $h = 6.63 \times 10^{-34} \text{ J s}$ .

Use  $K_{\text{max}} = h(f - f_0)$ ; frequency difference  $3.0 \times 10^{14} \text{ Hz}$ ;  $K_{\text{max}} = 1.99 \times 10^{-19} \text{ J}$ .

9702-22-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Nuclear physics?

Half-life is the time for half the undecayed nuclei to remain; decay is random for one nucleus but predictable statistically for a large sample.

9702-23-C2

QUESTION + ANSWER

Which is RMS voltage of a sinusoid with peak voltage  $V_0$ ?

- A  $V_0/\sqrt{2}$
- B  $\sqrt{2}V_0$
- C  $V_0/2$
- D  $2V_0$

A.  $V_0/\sqrt{2}$   
Why: An alternating voltage changes sign periodically.

9702-21-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Quantum physics?

Raising intensity increases emitted electron count at fixed frequency above threshold, while raising frequency increases maximum kinetic energy.

9702-22-C2

QUESTION + ANSWER

What is the central idea in Nuclear physics?

Nuclear mass defect is the difference between separate constituent masses and nuclear mass; binding energy is  $\Delta mc^2$ . Activity is decay events per second,  $A = \lambda N$ .

9702-23-C1

QUESTION + ANSWER

What is one becquerel?  
A One joule per second  
B One nucleus per kilogram  
C One decay per second  
D One half-life per second

C. One decay per second  
Why: Nuclear mass defect is the difference between separate constituent masses and nuclear mass; binding energy is  $\Delta mc^2$ .

9702-23-M

QUESTION + ANSWER

What is the central idea in Medical physics?

Ultrasound echoes locate boundaries because acoustic impedance mismatch reflects part of a pulse; depth equals speed times echo time divided by two. X-ray attenuation depends on material and thickness.

9702-24-C1

QUESTION + ANSWER

An echo returns  $120\ \mu\text{s}$  after emission; sound speed in tissue is  $1540\ \text{m s}^{-1}$ . Find depth and explain factor 2.

$d = vt/2 = 1540 \times 120 \times 10^{-6}/2 = 0.0924\ \text{m}$ ; measured time includes outward and return travel.

9702-24-P

QUESTION + ANSWER

If distance to a star doubles, its flux becomes what fraction?

- A One quarter
- B One half
- C Two times
- D Four times

A. One quarter

Why: Stellar luminosity is emitted power; observed flux falls as  $L/(4\pi d^2)$  for isotropic emission.

9702-25-M

QUESTION + ANSWER

A pendulum experiment records times for 10 oscillations. Explain why timing several oscillations improves the estimate of period, and name two controlled variables.

Longer total interval reduces reaction-time percentage error; control amplitude and bob/string setup or release method.

9702-26-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Medical physics?

CT reconstructs cross-sections from many projections; PET detects pairs of annihilation photons to map tracer distribution. Doses require benefit-risk assessment.

9702-24-C2

QUESTION + ANSWER

What is the central idea in Astronomy and cosmology?

Stellar luminosity is emitted power; observed flux falls as  $L/(4\pi d^2)$  for isotropic emission. A black-body star's peak wavelength decreases as temperature rises according to Wien's law.

9702-25-C1

QUESTION + ANSWER

A stellar spectrum peaks at  $500\ \text{nm}$ . Estimate surface temperature with Wien constant  $2.90 \times 10^{-3}\ \text{m K}$ .

$T = 2.90 \times 10^{-3}/(500 \times 10^{-9}) = 5800\ \text{K}$ .

9702-25-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

Calibrate instruments, record raw data with uncertainties and consistent precision, plot a graph with labelled axes and appropriate scale, then calculate gradient using a large triangle. Repeat measurements to identify anomalies.

9702-26-C2

QUESTION + ANSWER

Why is ultrasound echo time halved to calculate depth?

- A The probe halves frequency
- B The echo is absorbed twice
- C Speed halves in tissue
- D The sound makes a return trip

D. The sound makes a return trip

Why: Ultrasound echoes locate boundaries because acoustic impedance mismatch reflects part of a pulse; depth equals speed times echo time divided by two.

9702-24-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Astronomy and cosmology?

Cosmological redshift is evidence of expansion; for relatively nearby galaxies,  $v \approx H_0 d$ , with limitations at very large distances.

9702-25-C2

QUESTION + ANSWER

What is the central idea in P3 Advanced practical skills?

Plan to vary one independent variable while controlling others.

9702-26-C1

QUESTION + ANSWER

In a fair pendulum trial, which variable should remain controlled?

- A String length if studying length
- B Room temperature only
- C Number of pages in notes
- D Date of experiment

A. String length if studying length

Why: Plan to vary one independent variable while controlling others.

9702-26-M

QUESTION + ANSWER

FagusED · A-Level 9702  
P5 Planning, analysis and evaluation

What is the central idea in P5 Planning, analysis and evaluation?

State a testable relation, specify controls, propose apparatus and a safe repeatable method.

9702-27-C1

QUESTION + ANSWER

FagusED · A-Level 9702  
P5 Planning, analysis and evaluation

A model predicts  $F = kv^2$ . State graph axes to determine  $k$  and describe a check on experimental validity.

Plot  $F$  against  $v^2$ ; gradient gives  $k$ ; inspect straight-line fit, intercept and uncertainties/repeats.

9702-27-P

QUESTION + ANSWER

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P5 Planning, analysis and evaluation

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

Linearise a relationship to extract constants from gradient/intercept, propagate uncertainties and compare predicted values within those uncertainties. Improvements should target the dominant error mechanism.

9702-27-C2

QUESTION + ANSWER

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P5 Planning, analysis and evaluation

To test  $F = kv^2$ , what should be plotted against  $v^2$ ?

- A Force  $F$
- B Time  $t$
- C Mass  $m$
- D Velocity  $v$

A. Force  $F$

Why: State a testable relation, specify controls, propose apparatus and a safe repeatable method.

9702-27-M

QUESTION + ANSWER